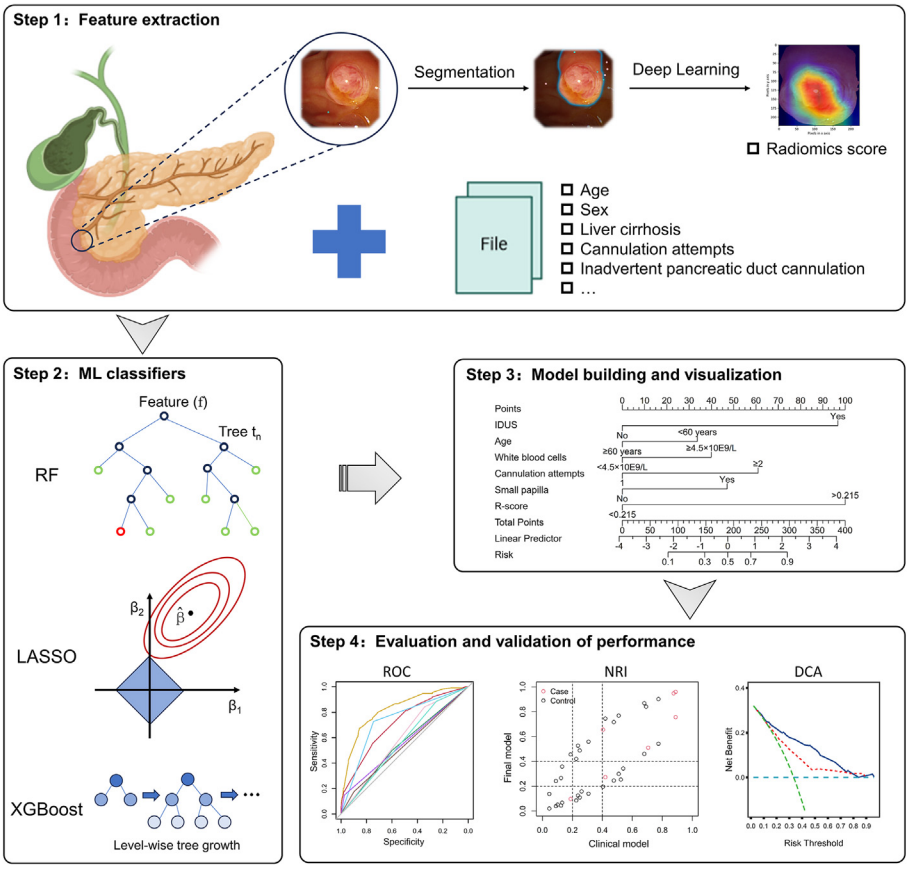
Duodenal papilla radiomics-based prediction model for post-ERCP pancreatitis using machine learning: a retrospective multicohort study

Kangjie Chen, PhD,[1,*] Haihao Lin, MD,[2,*] Feiyi Zhang, MD,[3] Ziying Chen, MD,[3] Huajie Ying, MD,[1] Linping Cao, PhD,[1] Jianfeng Fang, MD,[4] Danyang Zhu, MD,[5] Kewei Liang, PhD[2]

Hangzhou, Shaoxing, Haining, China

GRAPHICAL ABSTRACT



Background and Aims: The duodenal papillae are the primary and essential pathway for ERCP, greatly determining its complexity and outcome. We investigated the association between papilla morphology and post-ERCP pancreatitis (PEP) and constructed a robust model for PEP prediction.

Methods: We retrospectively enrolled patients who underwent ERCP in 2 centers from January 2019 to June 2022. Radiomic features of the papilla were extracted from endoscopic images with deep learning. Potential predictors and their importance were evaluated with 3 machine learning algorithms. A predictive model was developed using best subset selection by logistic regression, and its performance was evaluated in terms of discrimination, calibration, and clinical utility based on the area under curve (AUC) of the receiver-operating characteristic curve, calibration curve, and clinical decision curve, respectively.

Results: From 2 centers, 2038 and 334 ERCP patients were enrolled in this study with PEP rates of 7.9% and 9.6%, respectively. The radiomic score was significantly associated with PEP and showed great diagnostic value (AUC, .755-.821). Six hub predictors were selected to conduct a predictive model. The radiomics-based model demonstrated excellent discrimination (AUC, .825-.857) and therapeutic benefits in the training, testing, and validation cohorts. The addition of the radiomic score significantly improved the diagnostic accuracy of the predictive model (net reclassification improvement, .151-.583 [$P < .05$]; integrated discrimination improvement, .097-.235 [$P < .001$]).

Conclusions: The radiomic signature of the papilla is a crucial independent predictor of PEP. The papilla radiomics-based model performs well for the clinical prediction of PEP. (Gastrointest Endosc 2024;100:691-702.)

(footnotes appear on last page of article)

ERCP, initially introduced in 1968, has become an invaluable technique for diagnosing and treating numerous pancreaticobiliary disorders.[1] Post-ERCP pancreatitis (PEP) is the most serious adverse event of ERCP, with an incidence rate of 3.5% to 9.7% and a mortality rate of .1% to .7%.[1-3]

It is essential to understand the patient risk of PEP to facilitate early preventive treatment. Prophylactic pancreatic stents are considered the most effective preventive measure, reducing PEP risk by 60% to 80%, especially in severe cases.[4] However, their placement poses risks such as migration and bowel perforation, leading to increased patient discomfort and costs. Paradoxically, the 20% failure rate in placement attempts heightens the risk of PEP.[5] Hence, the development of a precise predictive model becomes crucial. Previous studies have identified various factors for PEP,[1,2] and some have established predictive models.[6-8] However, the predictive ability of these models is unsatisfactory or lacks validation.

The duodenal major papilla (DMP) is related to the confluence pattern of the pancreaticobiliary duct and greatly affects papillary cannulation.[9-11] Therefore, the relationship between papilla morphology and PEP is receiving more attention. Currently, DMP is commonly classified by the endoscopic appearance into 1 of 4 major papillary types: regular, small, protruding/pendulous, or creased/ridged.[11,12] The type of papillary orifice has been independently associated with PEP,[8] with small or ridged papilla associated with higher incidence rates.[13,14] However, the morphologic analysis of papilla images in these investigations is rather subjective, incomprehensible, and unquantified.

Deep learning has emerged as a powerful tool for autonomously learning representations of critical illness traits directly from photos to provide prediction information.[15,16] By building appropriate models with refined features, deep learning acquired successful assessment and prediction abilities in various challenging clinical tasks.[17-20]

This study aimed to develop and verify a predictive model for the occurrence of PEP based on large-scale clinical parameters and radiomic features derived from deep learning of papilla pictures, thereby minimizing the use of prophylactic pancreatic stents. In addition, the added predictive value of radiomics to clinical variables was explored.

METHODS

Study population

The study diagram is depicted in Figure 1. Between January 2019 and June 2022, 2038 patients underwent ERCP and were retrospectively enrolled at the First Affiliated Hospital, Zhejiang University School of Medicine (center 1). Exclusion criteria were age <18 years, prior ERCP history, pancreatic duct stent placement, acute pancreatitis within 3 days, abnormal duodenal papilla (conditions such as fistulas, impacted stones, tumors, or fish mouth–like openings because of intraductal papillary mucinous neoplasm, impairing normal function), surgically altered anatomy, biliary pancreatic duct confluence abnormality, and failed biliary cannulation.

Because the data of eligible patients were extremely unbalanced (PEP/non-PEP = 1:11.6), we performed propensity score matching (PSM) using 1:2 nearest-neighbor matching with a caliper of .05 based on indications, year of ERCP, and use of pancreatitis prophylactic drugs (eg, nonsteroidal anti-inflammatory drugs, somatostatin, or octreotide). Then, the PSM-adjusted data (derivation cohort) were randomly divided, with 70% for training and 30% for testing. For the validation cohort, patients were retrospectively enrolled from January 2022 to June 2022 at Shaoxing People's Hospital (center 2) and adjusted alike using PSM.

The Institutional Review Board of the Ethics Committee approved this study (IIT20230558A). The requirement for written informed consent was waived. All procedures involving human participants were performed in accordance with the 1975 Declaration of Helsinki and its later amendments.

ERCP procedure

All ERCP procedures were done using duodenoscopes (TJF-260V or JF-260V; Olympus, Tokyo, Japan), as described previously.[21] Briefly, after the papilla was reached and carefully examined, a wire-loaded sphincterotome (Dreamtome; Boston Scientific, Natick, Mass, USA) was initially used for selective

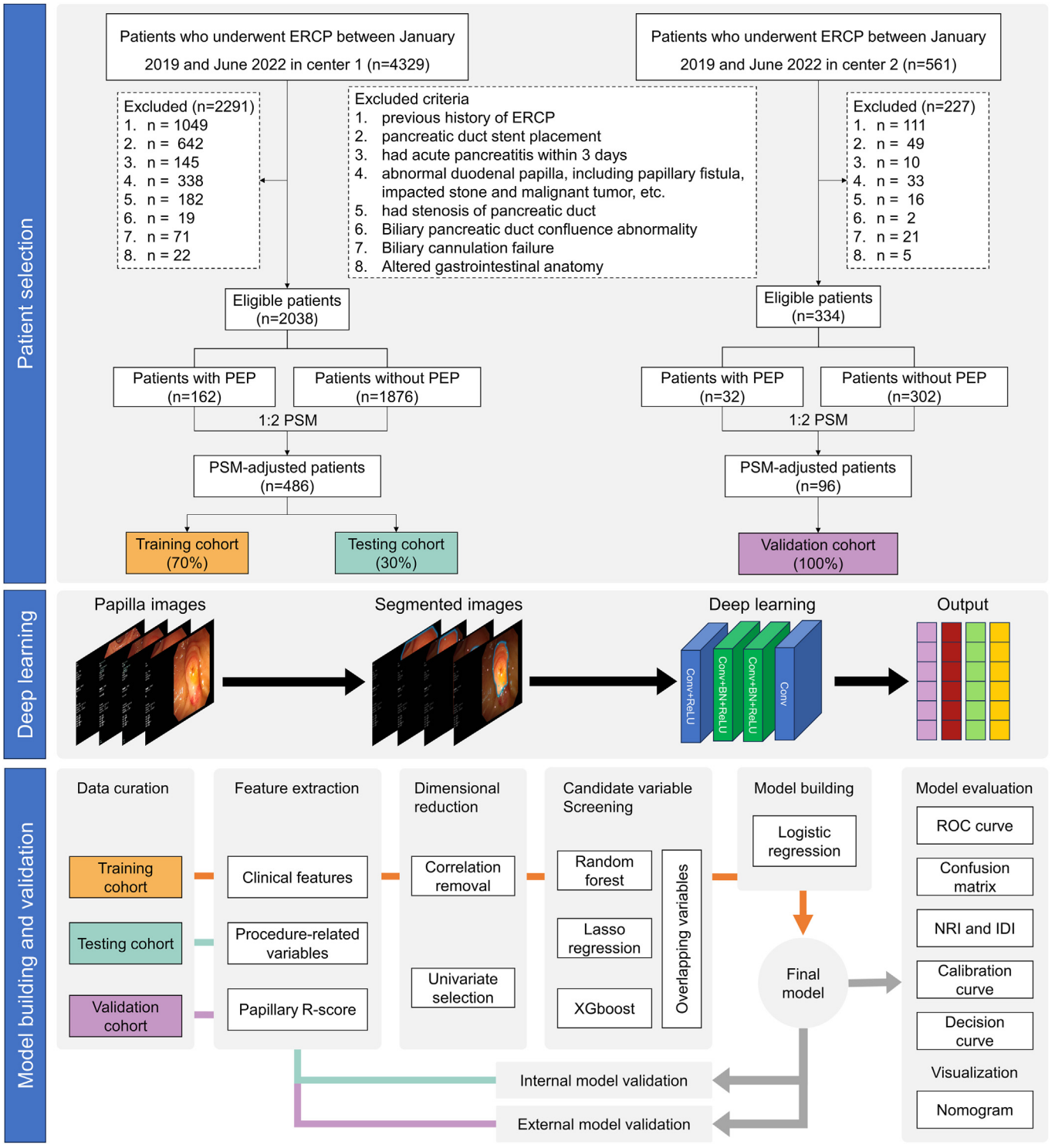


Figure 1. Overview of the study design. *PEP*, Post-ERCP pancreatitis; *PSM*, propensity score matching; *IDI*, integrated discrimination improvement; *Lasso*, least absolute shrinkage and selection operator; *NRI*, net reclassification improvement; *ROC*, receiver operator characteristic; *XGBoost*, eXtreme Gradient Boosting.

cannulation. If standard cannulation failed, a double-wire technique or precut (transpancreatic or needle-knife) was performed when appropriate. If trainees were involved in the procedure, 10 minutes was allowed to attempt cannulation. All other participating endoscopists had 15 years of experience in ERCP (>1000 per year). Therapeutic manipulation (eg, sphincterotomy, balloon dilation, stone extraction, and stent placement) was done when appropriate.

TABLE 1. Baseline demographic and ERCP-related characteristics in different cohorts

Variables*	Derivation cohort (n = 486)					
	Training cohort (n = 341)			Testing cohort (n = 145)		
	Non-PEP (n = 227)	PEP (n = 114)	*P* value	Non-PEP (n = 97)	PEP (n = 48)	*P* value
Clinical characteristics						
Sex, female	86 (38.0)	46 (40.0)	.659	41 (42)	23 (48)	.519
Age, y	60.33 ± 15.26	58.95 ± 15.30	.242	59.69 ± 15.44	59.27 ± 17.11	.936
Body mass index, kg/m^2	22.65 ± 3.02	23.20 ± 3.46	.359	22.90 ± 2.83	21.54 ± 3.43	.017
Pancreatic duct dilation‡	18 (7.9)	11 (9.6)	.591	8 (8.2)	4 (8.3)	.999
Coexisting disorders						
Hypertension	69 (30)	33 (29)	.783	24 (25)	11 (23)	.809
Diabetes	15 (6.6)	9 (7.9)	.661	7 (7.2)	2 (4.2)	.718
Liver cirrhosis	7 (3.1)	8 (7.0)	.095	4 (4.1)	1 (2.1)	.999
Indications			.606			.923
Cholelithiasis	105 (46)	50 (44)		35 (36)	20 (42)	
Malignant biliary stricture	79 (35)	41 (36)		41 (42)	19 (40)	
Benign biliary stricture	22 (9.7)	8 (7.0)		10 (10)	5 (10)	
Laboratory tests						
Alanine aminotransferase, U/L	147.72 ± 148.79	153.93 ± 168.15	.934	137.03 ± 127.40	129.13 ± 134.06	.484
γ-Glutamyl transferase, U/L	418.91 ± 421.96	397.74 ± 410.16	.581	388.02 ± 366.31	363.23 ± 374.27	.460
Albumin, g/L	40.53 ± 5.40	40.55 ± 5.31	.905	40.00 ± 5.97	39.93 ± 5.14	.635
Total bilirubin, μmol/L	95.50 ± 120.96	97.31 ± 119.16	.852	77.52 ± 103.02	99.14 ± 118.49	.483
Direct bilirubin, μmol/L	79.58 ± 105.92	81.49 ± 107.61	.717	59.74 ± 84.03	83.92 ± 105.65	.425
Serum calcium, mmol/L	2.24 ± .14	2.24 ± .14	.693	2.22 ± .16	2.25 ± .13	.260
Cholinesterase, U/L	6467.44 ± 2220.23	6465.08 ± 2227.01	.778	6542.09 ± 2294.96	5965.13 ± 2197.93	.196
White blood cell count, 10^9/L	6.57 ± 3.19	6.69 ± 2.74	.214	6.94 ± 3.85	6.16 ± 2.62	.188
Neutrophil ratio	66.43 ± 11.36	67.89 ± 11.49	.102	67.38 ± 10.73	64.65 ± 11.74	.193
Platelets, 10^9/L	220.30 ± 80.15	218.36 ± 77.81	.856	217.31 ± 96.87	217.73 ± 70.50	.465
Prothrombin time, s	11.95 ± 1.42	12.19 ± 2.16	.903	12.02 ± 1.98	11.79 ± .93	.832
Fibrinogen, g/L	3.72 ± 1.14	3.64 ± 1.05	.632	3.71 ± 1.29	3.27 ± 1.03	.043
International normal ratio	1.01 ± .11	1.01 ± .14	.853	1.01 ± .18	1.00 ± .08	.602

Values are n (%) or mean ± standard deviation.
PEP, Post-ERCP pancreatitis.
*Continuous variables used Wilcoxon rank sum test to observe intergroup difference, whereas categorical variables used the χ^2 test to observe intergroup difference.
†Statistical analyses were performed between the derivation cohort and validation cohort.
‡Pancreatic duct status, including dilation or stenosis, was assessed preoperatively through CT or MRCP.

Evaluation of the papilla, image segmentation, and deep learning model construction

We collected all endoscopic images of the ERCP included in this study as previously reported.[22] A volume of interest was segmented by 2 endoscopists (H.Y. and J.F., with 10 years of experience in ERCP and over 1000 ERCP procedures performed) who were blinded to the clinical and pathologic data using Segment Anything (https://segment-anything.com).[23] All images were segmented into 3 regions (entire image, papilla, and orifice) and then reviewed together by the 2 readers after the individual evaluation. Any disagreement between the readers was discussed until a final consensus was generated. The deep learning architecture was composed using Python 3.9.7 (Python Software Foundation, Wilmington, Del, USA) and the open-source PyTorch 1.10 (Facebook AI Research lab, Menlo Park, Calif, USA) library. The model output was converted into a probability score, namely the radiomic score (R-score), indicating the individual relative risk for PEP.[17] More detailed information is provided in Supplementary Figure 1 (available online at www.giejournal.org).

Data collection

During the performance of ERCP, data related to cannulation procedure were prospectively collected by 1 investigator from each center, including cannulation method, cannulation

TABLE 1. Continued

Validation cohort (n = 96)			
Non-PEP (n = 64)	PEP (n = 32)	*P* value	*P* value†
20 (31)	10 (31)	>.999	.095
59.64 ± 14.09	59.00 ± 14.33	.807	.584
22.35 ± 2.95	23.49 ± 2.59	.064	.843
6 (9.4)	2 (6.3)	.715	.974
43 (67)	23 (72)	.640	.545
6 (9.4)	1 (3.1)	.418	.859
5 (7.8)	2 (6.3)	>.999	.185
		>.999	.240
30 (47)	15 (47)		
24 (38)	12 (38)		
2 (3.1)	1 (3.1)		
138.39 ± 183.13	149.64 ± 146.43	.371	.349
465.98 ± 586.09	399.08 ± 368.77	.712	.804
38.15 ± 6.60	42.04 ± 4.93	.008	.269
104.98 ± 138.88	46.86 ± 49.85	.110	.667
92.38 ± 126.77	36.31 ± 48.10	.052	.838
2.25 ± .16	2.30 ± .13	.150	.083
5610.14 ± 2204.74	6671.19 ± 1941.87	.026	.066
6.80 ± 3.14	6.38 ± 2.51	.611	.822
68.60 ± 11.73	87.54 ± 115.57	.718	.144
201.03 ± 78.13	218.72 ± 77.88	.256	.219
12.57 ± 1.86	11.65 ± 2.25	.182	.007
3.81 ± 1.18	4.14 ± 1.55	.529	.085
1.08 ± .17	1.02 ± .09	.221	<.001

time, number of cannulation attempts, and inadvertent pancreatic duct cannulation. Definitions, cutoffs, and scores of candidate factors are listed in Supplementary Table 1 (available online at www.giejournal.org). After ERCP, all patients were followed up to assess immediate and delayed serious adverse events until discharge. The diagnosis of PEP was based on the Atlanta criteria.[24] Serious adverse events were recorded according to published guidelines.[2]

Development and validation of the predictive model

The model development and optimization work were based on a multi-stepwise pipeline (Fig. 1). First, variables were filtered by univariable selection (variables with $P < .2$ were kept), and the Spearman correlation was calculated to assess their multi-collinearity. For coefficient values $\geq .7$ or less than or equal to $-.7$ for a pair of variables, only the variable with better diagnostic performance was retained.[25] Second, we fed candidate variables into 3 machine learning (ML) algorithms in a parallel manner: the least absolute shrinkage and selection operator (LASSO) penalized regression, random forest (RF), and eXtreme Gradient Boosting (XGBoost). Detailed information about the ML process can be found Appendix 1 (available online at www.giejournal.org). Each algorithm was trained to accurately fit the labeled data and predict the testing cohorts. A 10-fold cross-validation scheme was used to optimize model hyperparameters. Finally, the intersected predictors identified by ML algorithms were taken to fit multivariate logistic regression to develop a model with the lowest Akaike information criteria and tested in the validation cohort.

The receiver-operating characteristic (ROC) curve, calibration curve, and decision curve were used to demonstrate the prediction ability of models visually. The diagnostic

indexes of models, including area under curve (AUC), specificity, sensitivity, accuracy, confusion matrix, net reclassification improvement (NRI), and integrated discrimination improvement (IDI), were also calculated.

Statistical analysis

Statistical analysis was conducted with R software (version 3.0.1, The R Foundation for Statistical Computing, Vienna, Austria). Continuous data are presented as mean ± standard deviation and categorical variables as number of subjects (with percentages). The cutoff of continuous data was determined by the maximal Youden index. All P values were obtained in 2-tailed tests, and $P < .05$ was considered statistically significant.

RESULTS

Study population and baseline characteristics

Between January 2019 and June 2022, 2038 and 334 ERCP patients were enrolled in center 1 and center 2, respectively. The PEP incidence in center 1 was lower than that in center 2, but the difference was not significant (7.9% [162/2038] vs 9.6% [32/334], $P = .313$) (Fig. 1). After PSM, 486 patients were selected for the derivation cohort and 96 patients for the validation cohort (Fig. 1 and Table 1). Between the 2 cohorts, no significant differences were observed in age, sex, body mass index, laboratory tests, or coexisting disorders, except for significant differences in prothrombin time ($P = .007$) and international normal ratio ($P < .001$). Moreover, baseline characteristics were similar between patients with and without PEP in all cohorts (Table 1).

Radiomic analysis

We trained a deep neural network to predict PEP probability using associations of papillary features. The software developed different approaches to analyze the different regions of the DMP, which were carefully selected and quantitatively integrated radiomic features into 12 R-score groups (Fig. 1 and Supplementary Fig. 1). Among them, the R-score of the papilla generated by the model ResNet50 (Microsoft Research, Redmond, Wash, USA) yielded the highest AUC value to achieve statistical significance (for most, $P < .05$) (Supplementary Table 2, available online at www.giejournal.org). By applying the ResNet50 model to predict PEP based on papilla features, clear differences in the R-score were noted ($P < .001$) (Fig. 2A) between the PEP and non-PEP groups with powerful diagnostic performance (AUC, .755-.821; $P < .001$ for R-score vs papillary type in each cohort) (Fig. 2B). We further assessed the predictive accuracy of the R-score for diverse clinical outcomes, including cannulation difficulty, guidewire technique, and the need for precut sphincterotomy, but the results were unsatisfactory (Supplementary Fig. 2, available online at www.giejournal.org).

Notably, R-scores were similar between different papillary types when classified endoscopically (Fig. 2C). Next, we used probability density plots for PEP and non-PEP patients to denote the optimal separation margins for risk prediction (Fig. 2D). The cutoff value of the R-score was determined to be .215, yielding excellent separation between classes ($\chi^2 = 69.88$, $P < .001$) (Fig. 2E). Correlation analysis showed no strong correlation between R-score and other variables with correlation coefficient values between −.01 and .13 (Fig. 2F).

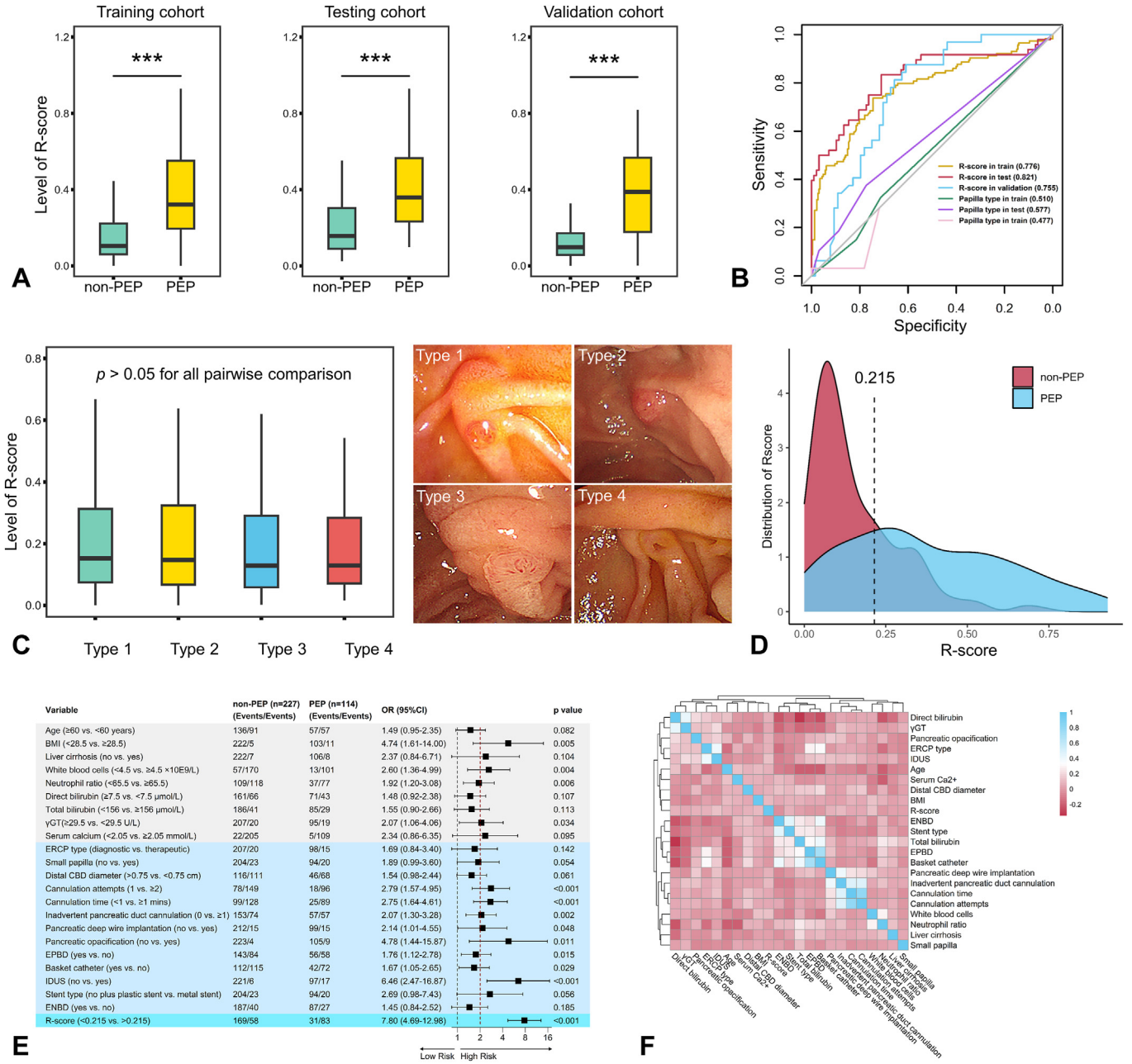
Identification of potential predictors by ML algorithms

Among 49 included variables, 9 patient-related variables, 13 procedure-related variables, and R-score were identified in univariable analysis (all $P < .2$) (Fig. 2E and Supplementary Fig. 3, available online at www.giejournal.org). The Spearman correlation analysis showed a high correlation of cannulation attempts with cannulation time ($r = .79, P < .001$) and endoscopic papillary balloon dilatation with basket catheters ($r = .72, P < .001$) in the training cohort (Fig. 2F). Therefore, cannulation time and basket catheter were removed.

To further decrease the variable dimension and simplify the model, we used 3 ML algorithms acting in parallel. After hyperparameter optimization (Fig. 3A-D), the top 10 important variables were identified based on the regression coefficient (Fig. 3E), the mean decrease in Gini coefficient (Fig. 3F), and the Shapley Additive Explanations value (Fig. 3G) in the 3 algorithms, respectively. The ROC analysis in the test cohort showed similar performances for the LASSO regression (AUC, .827; 95% confidence interval [CI], .751-.902) and RF algorithms (AUC, .781; 95% CI, .698-.864) and XGBoost algorithms (AUC, .789; 95% CI, .704-.873) (Fig. 3H). The intersecting variables of the 3 algorithms were selected through a Venn diagram, which identified 6 variables (age, white blood cell count, cannulation attempts, small papilla, intraductal US [IDUS], and R-score) as the critical predictors (Fig. 3I).

Model establishment by logistic analysis

The 6 variables noted above were further assessed by multivariate logistic regression in the training cohort. This analysis demonstrated each factor to be an independent predictor for PEP, with R-score having the greatest impact, as follows: age (odds ratio [OR], 2.00; 95% CI, 1.14-3.48; $P = .015$), white blood cell count (OR, 2.27; 95% CI, 1.09-4.74; $P = .029$), small papilla (OR, 2.63; 95% CI, 1.20-5.76; $P = .016$), cannulation attempts (OR, 3.49; 95% CI, 1.79-6.82; $P < .001$), IDUS (OR, 7.28; 95% CI, 2.20-24.06; $P = .001$), and R-score (OR, 7.79; 95% CI, 4.48-13.55; $P < .001$) (Fig. 4A). These independent factors were used to form a predictive model, described by the formula $Y_{\text{final model}} = -3.886 + .691 \times \text{age} + .820 \times \text{white blood cell count} + .965 \times \text{small papilla} + 1.251 \times \text{cannulation attempts} + 1.986 \times \text{IDUS} + 2.053 \times \text{R-score}$. This model was presented as the nomogram (Fig. 4B and Supplementary Fig. 4, available online at www.



Variable	non-PEP (n=227) (Events/Events)	PEP (n=114) (Events/Events)	OR (95%CI)	p value
Age (≥60 vs. <60 years)	136/91	57/57	1.49 (0.95-2.35)	0.082
BMI (<28.5 vs. ≥28.5)	222/5	103/11	4.74 (1.61-14.00)	0.005
Liver cirrhosis (no vs. yes)	222/7	106/8	2.37 (0.84-6.71)	0.104
White blood cells (<4.5 vs. ≥4.5 ×10E9/L)	57/170	13/101	2.60 (1.36-4.99)	0.004
Neutrophil ratio (<65.5 vs. ≥65.5)	109/118	37/77	1.92 (1.20-3.08)	0.006
Direct bilirubin (≥7.5 vs. <7.5 μmol/L)	161/66	71/43	1.48 (0.92-2.38)	0.107
Total bilirubin (<156 vs. ≥156 μmol/L)	186/41	85/29	1.55 (0.90-2.66)	0.113
γGT(≥29.5 vs. <29.5 U/L)	207/20	95/19	2.07 (1.06-4.06)	0.034
Serum calcium (<2.05 vs. ≥2.05 mmol/L)	22/205	5/109	2.34 (0.86-6.35)	0.095
ERCP type (diagnostic vs. therapeutic)	207/20	98/15	1.69 (0.84-3.40)	0.142
Small papilla (no vs. yes)	204/23	94/20	1.89 (0.99-3.60)	0.054
Distal CBD diameter (>0.75 vs. <0.75 cm)	116/111	46/68	1.54 (0.98-2.44)	0.061
Cannulation attempts (1 vs. ≥2)	78/149	18/96	2.79 (1.57-4.95)	<0.001
Cannulation time (<1 vs. ≥1 mins)	99/128	25/89	2.75 (1.64-4.61)	<0.001
Inadvertent pancreatic duct cannulation (0 vs. ≥1)	153/74	57/57	2.07 (1.30-3.28)	0.002
Pancreatic deep wire implantation (no vs. yes)	212/15	99/15	2.14 (1.01-4.55)	0.048
Pancreatic opacification (no vs. yes)	223/4	105/9	4.78 (1.44-15.87)	0.011
EPBD (yes vs. no)	143/84	56/58	1.76 (1.12-2.78)	0.015
Basket catheter (yes vs. no)	112/115	42/72	1.67 (1.05-2.65)	0.029
IDUS (no vs. yes)	221/6	97/17	6.46 (2.47-16.87)	<0.001
Stent type (no plus plastic stent vs. metal stent)	204/23	94/20	2.69 (0.98-7.43)	0.056
ENBD (yes vs. no)	187/40	87/27	1.45 (0.84-2.52)	0.185
R-score (<0.215 vs. >0.215)	169/58	31/83	7.80 (4.69-12.98)	<0.001

Figure 2. Radiomic analysis of the duodenal major papilla. **A,** Box plots of radiomic score (R-score) levels between post-ERCP pancreatitis (PEP) and non-PEP groups. The *box* bounds the interquartile range divided by the median, with the *whiskers* extending to a maximum of 1.5 times the interquartile range beyond the box. ***$P < .001$. **B,** Receiver-operating characteristic curves for the R-score and papillary type in the training, testing, and validation cohorts. **C,** Box plots depicting R-score variations across papillary types, shown by images as type 1, regular; type 2, small; type 3, protruding/pendulous; and type 4, creased/ridged. **D,** Probability density distributions of R-scores for PEP and non-PEP categories. The *dotted line* indicates the optimal cutoff. **E,** Forest plot presenting ORs, CIs, and *P* values for patient-related (*gray background*) and procedure-related (*light blue background*) predictors and R-scores between the non-PEP and PEP groups. **F,** Heatmap visualizing Spearman correlations among various clinical variables. *OR*, Odds ratio; *CI*, confidence interval; *BMI*, body mass index; *γGT*, γ-glutamyl transferase; *CBD*, common bile duct; *EPBD*, endoscopic papillary balloon dilatation; *IDUS*, intraductal US; *ENBD*, endoscopic nasobiliary drainage.

giejournal.org) and displayed good accuracy in estimating the risk of PEP, with an AUC of .825 (95% CI, .778-.872) in the training cohort, .836 (95% CI, .769-.903) in the testing cohort, and .857 (95% CI, .770-.945) in the validation cohort (Fig. 4C). The confusion matrix from this model showed a true positive rate of >80% for all cohorts (84%, 86%, and 89% in the training, testing, and validation cohorts, respectively) (Fig. 4D). The Hosmer-Lemeshow test showed a nonsignificant statistic in the training ($P = .951$), testing ($P = .332$), and validation ($P = .112$) cohorts (Fig. 4E), which indicates excellent calibration of the model across all cohorts.

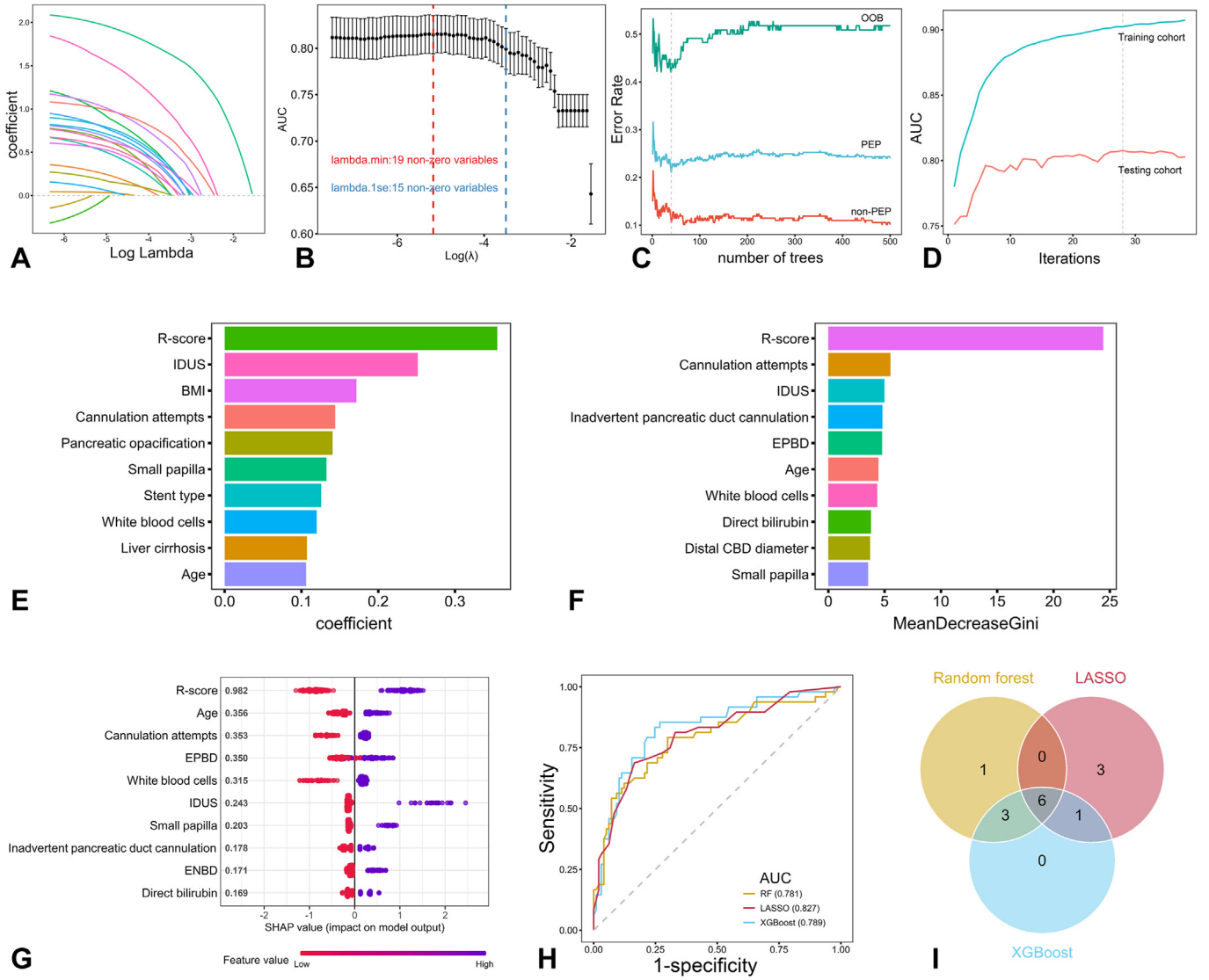


Figure 3. Selection of dominant predictors by machine learning algorithms. **A,** Least absolute shrinkage and selection operator (LASSO) coefficient profiles of the 20 potential predictors. **B,** Determination of the optimal tuning parameter (λ) in the LASSO binary logistic regression model. Selection was based on 10-fold cross-validation using the minimum criteria. **C,** Determination of optimal hyperparameters for the random forest (RF) model. The optimal number of trees is 40 for the testing cohort (*gray dotted line*). **D,** The AUC over iterations in the , eXtreme Gradient Boosting (XGBoost) model for the training (*green line*) and testing (*red line*) cohorts. The optimal number of iterations is 28 for the testing cohort (*gray dotted line*). **E-G,** The top 10 important variables ranked according to the value of the regression coefficient, the mean decrease in Gini coefficient, and Shapley Additive Explanations in the LASSO (**E**), RF (**F**), and XGBoost (**G**) algorithms, respectively. The R-score is the top-ranked factor in all algorithms. **H,** Receiver-operating characteristic curves of LASSO, RF, and XGBoost models in the testing cohort. **I,** The final 6 hub predictors were filtered out by the 3 algorithms in the derivation cohorts. *BMI*, Body mass index; *γGT*, γ-glutamyl transferase; *CBD*, common bile duct; *EPBD*, endoscopic papillary balloon dilatation; *IDUS*, intraductal US; *ENBD*, endoscopic nasobiliary drainage; *R-score*, radiomic score; *AUC*, area under the curve; *OOB*, out-of-bag; *PEP*, post-ERCP pancreatitis.

Incremental predictive ability of the radiomics-based model

Using an optimal threshold chosen to maximize the Youden index of the ROC analysis from the training cohort, we summarized the diagnostic performance of models and independent predictors (Fig. 5A). The Final model had the highest accuracy (>75%) for PEP prediction. Compared with the clinical model (composed of 5 independent predictors other than the R-score), the final model had statistically significant and positive NRI and IDI compared with clinical models in the training (NRI: .583 [95% CI, .407-.753], $P < .001$; IDI: .235 [95% CI, .187-.282], $P < .001$), testing (NRI: .247 [95% CI, .006-.489], $P < .001$; IDI: .097 [95% CI, .048-.147], $P < .001$), and validation (NRI: .172 [95% CI, .096-.440], $P < .05$; IDI: .151 [95% CI, .079-.223], $P < .001$) cohorts (Fig. 5B). These results indicated that the final model had better reclassification of PEP and non-PEP cases as well as a higher integrated sensitivity and integrated specificity than the clinical model. The AUCs of the final model were

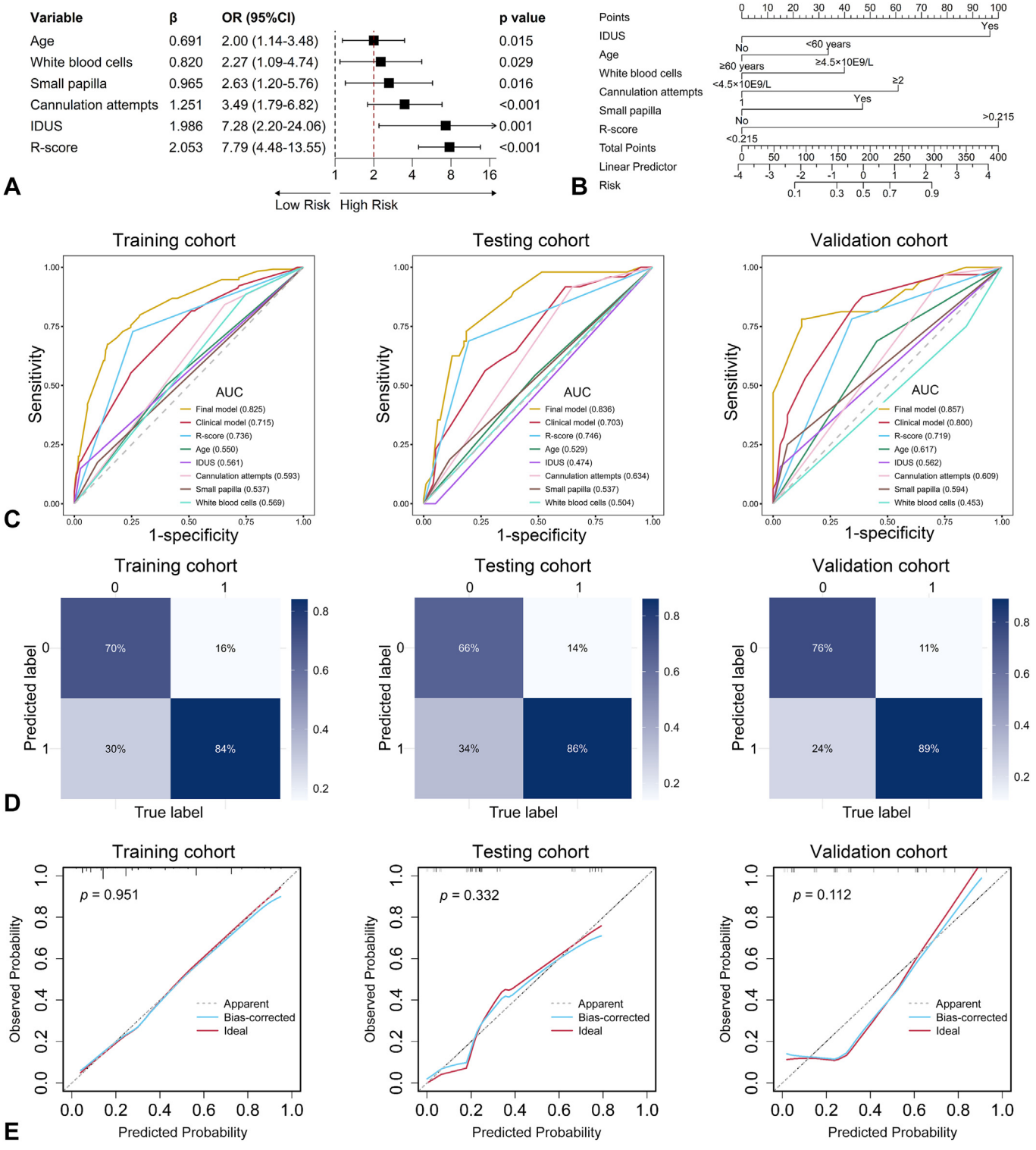


Figure 4. Evaluation of model performance. **A,** Forest plot of independent predictors of post-ERCP pancreatitis with a multivariate logistic regression model. **B,** The model presented with a nomogram scaled by the proportional regression coefficient of each predictor. **C,** Plots of receiver operating characteristic curves for predictive models and independent risk factors in the training, testing, and validation cohorts. **D,** Confusion matrix of the final model for the training, testing, and validation cohorts. The color intensity within each cell corresponds to the percentage value, with *darker shades* representing higher values. **E,** Calibration curves of the final model in the training, testing, and validation cohorts. The curves illustrate the observed versus predicted probabilities. The *Ideal line* denotes perfect calibration, the *Apparent line* shows the model's performance, and the *Bias-corrected line* indicates the performance after accounting for over-optimism. *OR*, Odds ratio; *CI*, confidence interval; *IDUS*, intraductal US; *R-score*, radiomic score; *AUC*, area under the curve.

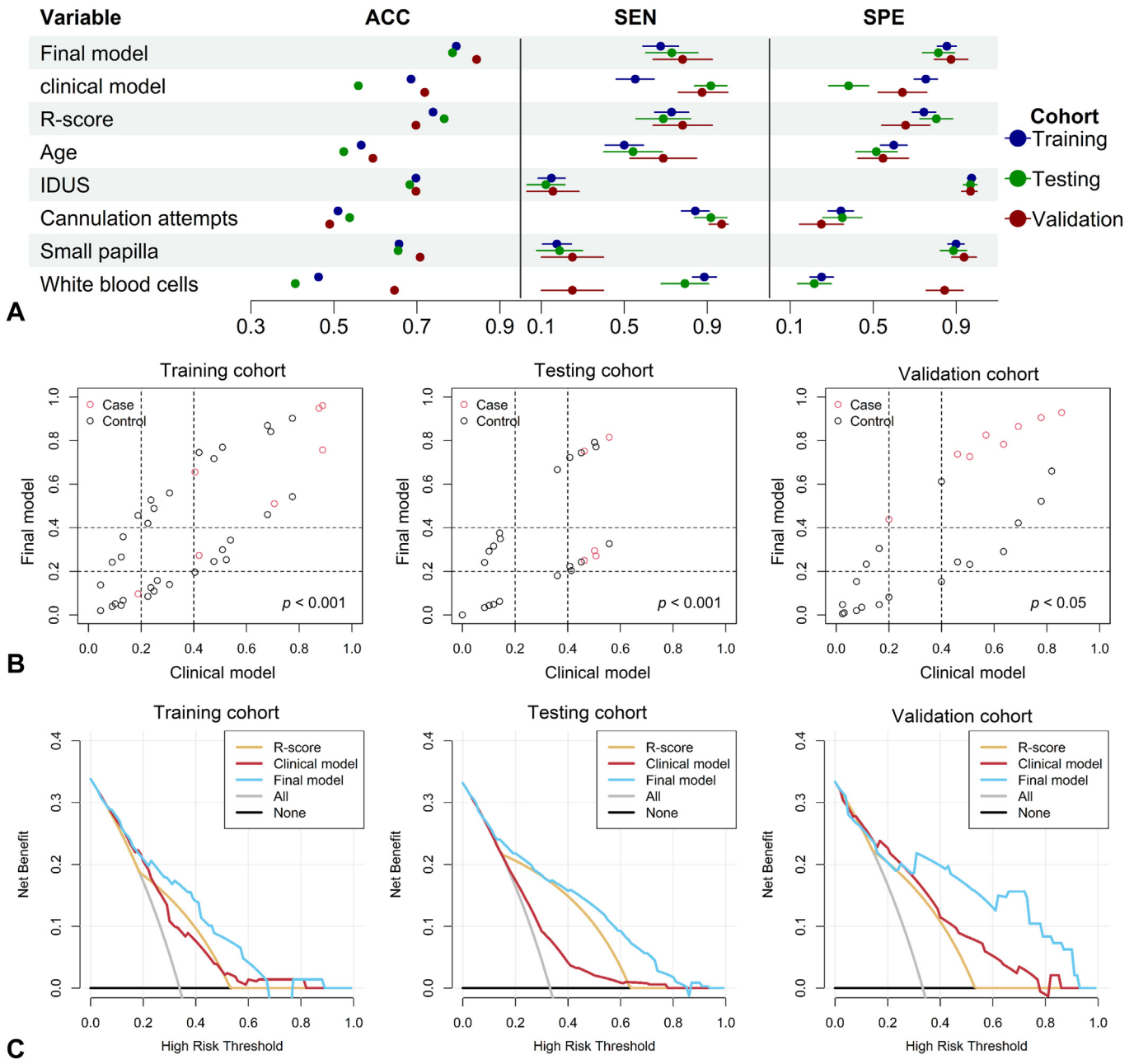


Figure 5. Comparison of diagnostic performance between clinical and radiomics-based models across different cohorts. **A,** Forest plot illustrating the accuracy, sensitivity, and specificity of the final model, clinical model, and independent variables across the training, testing, and validation cohorts. Each *point* on the plot represents the metric value for the respective model or variable in a specific cohort, with *horizontal lines* indicating the 95% confidence interval. **B,** Net reclassification index analysis between the final and clinical models. **C,** Decision curve analyses of the R-score, clinical model, and Final model for the 3 cohorts. *ACC*, Accuracy; *SEN*, sensitivity; *SPE*, specificity; *IDUS*, intraductal US; *R-score*, radiomic score.

higher than those of the clinical model in each cohort, although the difference was not significant in the validation cohort (Fig. 4C).

To evaluate our model's clinical value, we used a decision curve analysis to identify which model score interval could most accurately predict PEP before cannulation. This analysis demonstrated that the use of a radiomics-based model provided superior PEP prediction over the "treat all" or "treat none" schemes when the threshold probability exceeded 10% (Fig. 5C). Within this threshold, the radiomics-based model showed greater clinical utility than the clinical model across all cohorts.

Predictive ability for moderate to severe PEP

Patients with moderate to severe pancreatitis showed significant differences in both hospital duration and postoperative amylase levels compared with their mild counterparts ($P <$ .001), with a marked increase in the mortality rate ($P <$ .001)

(Supplementary Table 3, available online at www.giejournal.org). This underscores notably poorer clinical outcomes for patients with moderate to severe pancreatitis. The predictive model exhibited significantly higher AUC values than the R-score (.775 vs .697 [$P < .001$] in the derivation cohort and .780 vs .594 [$P = .003$] in the validation cohort) (Supplementary Table 4, available online at www.giejournal.org), underscoring its potential advantage in accurately predicting moderate to severe PEP.

DISCUSSION

In this study, we developed a papilla morphology–based radiomics signature by deep learning analysis of endoscopic images for PEP prediction. By using 3 distinct ML algorithms (LASSO, RF, and XGBoost), we identified 6 critical predictors of PEP, with the radiomics signature being the most important. The fully integrated model incorporating the radiomics signature outperformed the clinical model in terms of calibration and discriminative ability and further provided significant net benefits in clinical settings.

The pathogenesis of PEP is complex and is probably the result of the combination of chemical, hydrostatic, enzymatic, mechanical, and thermal insults that occur during ERCP.[26] PEP-related risk factors are usually divided into 3 types: patient-related, procedure-related, and operator-related factors.[1,2] Pharmacologic measures can be used to prevent severe PEP if it can be predicted in advance. However, a single reliable factor to predict PEP is still lacking. Papilla-related factors, such as the size, opening, morphology, and location of the DMP, have been implicated in the risk of PEP.[11-14,27] Wang et al[22] showed that a long- to short-axis ratio of DMP of $\geq$1.5 and the absence of periampullary diverticulum were independent high-risk factors for PEP. Our study also identified small papilla to be an independent risk factor for PEP (OR, 2.63; 95% CI, 1.20-5.76; $P = .016$).

In a significant shift from previous studies, we used a deep learning approach for the radiomic analysis of DMP. This innovative methodology underlines the critical role of artificial intelligence in the extraction and analysis of radiomic imaging features, enhancing our ability to understand and predict complex biologic phenomena. Radiomics can be used to capture tissue properties such as shape, heterogeneity, and—on serial imaging—subtle changes over time.[28] This enables comprehensive and quantitative analysis of volume of interest and minimizes the subjective influence of observers. Encouragingly, we found that the R-score, derived from radiomic analysis, was the highest ranking variable in ML algorithms (Fig. 3E and F) and the foremost predictor for PEP in multivariate analysis (OR, 7.79; 95% CI, 4.48-13.55; $P < .001$) (Fig. 4A). It demonstrated a powerful diagnostic performance with significantly higher AUC values than those of the other independent predictors in each cohort (all $P < .001$) (Fig. 4C).

Other independent predictors in our study were age, white blood cell count, cannulation attempts, and IDUS, different from those described by the American Society for Gastrointestinal Endoscopy and European Society of Gastrointestinal Endoscopy.[1,2] For example, younger age (<35 years) and IDUS are "likely" risk factors rather than "definite" risk factors for PEP in the European Society of Gastrointestinal Endoscopy guidelines.[2] However, our results demonstrated a higher incidence of PEP among patients aged <60 years (OR, 2.00; 95% CI, 1.14-3.48; $P = .015$) (Fig. 4A), and IDUS was an important factor after R-score (OR, 7.28; 95% CI, 2.20-24.06; $P = .001$) (Fig. 4A). The discrepancy may be attributed to the following reasons: first, we specifically chose patients with a naive papilla and excluded patients who underwent prophylactic pancreatic stent placement; and second, we used multiple ML methods to identify risk factors, which helped mitigate bias and confounding factors while avoiding overfitting.

Currently, several models have been built to predict the presence of PEP based on patient- and procedure-related risk factors identified by univariate and multivariate analyses.[6-8] Chiba et al[6] established a simplified scoring system using 5 procedure-related factors with an AUC of .81. He et al[7] constructed a model based on 3 patient-related factors and 2 procedure-related factors and achieved an AUC of .676. The major limitations of these models were low credibility because of a too-small sample size, lack of validation, and impracticable complexity. On the other hand, because of the advancement of endoscopic techniques (eg, multilateral metal stent placement) and devices (eg, single-operator cholangioscopy), the existing scoring criteria need to be updated using a recent clinical and procedural database. Finally, these models predominantly rely on conventional methodology. Hence, they fall short in capitalizing on the predictive potential harbored within the wealth of radiomic imaging data. Through the power of deep learning, we were able to discern subtle patterns in the imaging data, significantly enhancing the predictive accuracy of our models. The addition of the R-score to the predictive model significantly increased the diagnostic performance in the training (AUC, .715 vs .825; $P = .011$), testing (AUC, .715 vs .825; $P = .012$) and validation (AUC, .715 vs .825; $P = .112$) cohorts (Fig. 4C). These results indicate that the radiomics-based model has better predictive values for PEP. Furthermore, the model exhibited greater clinical utility, as indicated by the higher net benefits in the decision curve analysis (Fig. 5C).

Some limitations of this study should also be addressed. First, our risk prediction models were developed using a retrospective cohort primarily from a single center, impacting our statistical analysis and generalizability. Second, many patients who met the inclusion criteria were ultimately excluded because of missing data or poor visualization of the DMP. Potentially, this could cause a selection bias, hampering the reproducibility and comparability of our results. These limitations would be mitigated with a prospective, multicenter study with larger sample sizes, which would help to validate this model and refine its prediction ability.

In conclusion, a nomogram incorporating patient-related factors, procedure-related factors, and DMP-based radiomics signature had a robust performance in predicting the PEP. These results suggest this model could play a possible role as an adjunctive method in the clinical practice and decision-making after ERCP.

DISCLOSURE

All authors disclosed no financial relationships.

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Abbreviations: AUC, area under curve; CI, confidence interval; DMP, duodenal major papilla; IDI, integrated discrimination improvement; IDUS, intraductal US; LASSO, least absolute shrinkage and selection operator; ML, machine learning; NRI, net reclassification improvement; OR, odds ratio; PEP, post-ERCP pancreatitis; PSM, propensity score matching; RF, random forest; ROC, receiver-operating characteristic; R-score, radiomic score; XGBoost, eXtreme Gradient Boosting.

*Drs Chen and Lin contributed equally to this article.

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Current affiliations: Department of Hepatobiliary and Pancreatic Surgery, the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China (1), School of Mathematical Sciences (2), Polytechnic Institute (3), Zhejiang University, Hangzhou, China; Division of Hepatobiliary Pancreatic Surgery, Department of Surgery, Shaoxing People's Hospital, Shaoxing, China (4), Division of Oncological Surgery, Department of Surgery, Haining Hospital of Traditional Chinese Medicine, Haining Cancer Hospital, Haining, China (5).

Reprint requests: Kewei Liang, PhD, School of Mathematical Sciences, Zhejiang University, 866 Yuhangtang Rd, Xihu District, Hangzhou 310058, China.

APPENDIX 1

Evaluation of the papilla

We collected all endoscopic images from the ERCP included in this study as previously reported.[1] Briefly, the duodenal major papilla (DMP) was carefully evaluated using a duodenoscope (JF-260V or TJF-260V; Olympus Optical, Tokyo, Japan). When the duodenoscope was advanced into the descending duodenum, a short position was preferred. By adjusting the position of the duodenoscope, the opening of the DMP was placed in the center or lower part of the screen. The long axis of the DMP was placed in the "twelve o'clock" position if possible. Three pictures with a good view of the entire papilla (eg, orifice, upper end, and transverse folds) and surrounding duodenal wall (especially the periampullary diverticulum) were taken for each papilla. If the papilla could not be fully exposed for an optimal picture, a sphincterotome was used to assist the exposure while avoiding distorting its morphology.

Image segmentation and deep learning model construction

For duodenal papilla segmentation, a volume of interest (VOI) was delineated by 2 endoscopists (H.Y. and J.F., with 10 years of experience in ERCP and over 1000 ERCP procedure volumes) who were blinded to the clinical and pathologic data with an online software deposited on the Segment Anything website (https://segment-anything.com).[2] All images were segmented into 3 regions (entire image, papilla, and orifice) and then reviewed together by the 2 readers after the individual evaluation (Supplementary Fig. 1). Any disagreement between the readers was discussed until a final consensus was generated.

In the subsequent deep learning analysis, the smallest square bounding box encapsulating the VOI was used as the input. We resized all VOIs to a fixed size of 224 × 224. Augmentation was performed in the training set to increase the samples. The ResNet50 (Microsoft Research) architecture, as deployed in this study, is depicted in Supplementary Figure 1. The training phase used Stochastic Gradient Descent as the optimizer, in conjunction with a Cosine Annealing Scheduler to adjust the learning rate dynamically.[4] The initial learning rate for the optimizer was established at .0001, and the decay range was set at (1 – .01). The following formula was used to compute the learning rate for the Cosine Annealing Scheduler:

$$\mathrm{lr} = \frac{1 + \cos\left(\frac{x \cdot \pi}{\mathrm{epochs}}\right)}{2} \cdot (1 - \mathrm{lrf}) + \mathrm{lrf}$$

where x signifies the current epoch, epochs corresponds to the total number of epochs, and lrf denotes the minimum learning rate fraction.

The study used the rectified linear unit as the activation function and cross-entropy loss as the loss function. The training process consisted of 300 iterations with a batch size of 8 and was terminated prematurely if validation accuracy plateaued to prevent overfitting. The optimal model was determined based on the highest accuracy achieved on the validation set, which was evaluated using the original image data.

The performance of the classification was assessed by calculating the accuracy on the validation set, with a distribution of 80% of cases for training and the remaining 20% for testing. During validation, only the VOI was inputted without executing the data augmentation operations mentioned previously. At the end of the iterations, the output of model was converted into a probability score, namely the radiomic score, indicating the individual relative risk for post-ERCP pancreatitis. The software code was composed using Python 3.9.7 and the open-source PyTorch 1.10 library (under the Berkeley Software Distribution 3-Clause license) and executed on a Graphics Processing Unit-optimized workstation featuring a single NVIDIA GeForce GTX 3090 (NVIDIA Corporation, Santa Clara, Calif, USA) (24GB, Ampere architecture).[3]

Machine learning algorithms

Least absolute shrinkage and selection operator. The least absolute shrinkage and selection operator (LASSO) is a linear model that estimates sparse coefficients. It is useful in some contexts because of its tendency to prefer solutions with fewer non-zero coefficients, effectively reducing the number of features on which the given solution depends. For this reason, LASSO and its variants are fundamental to the field of compressed sensing. Under certain conditions, it can recover the exact set of non-zero coefficients. Because the LASSO regression yields sparse models, it can thus be used to perform feature selection, as detailed in L1-based feature selection.

Given a dataset of n samples (x_1, y_1), (x_2, y_2), …, $(x_n, (y_n))$, where $x_i \in \mathbb{R}^p$ and $y_i \in \mathbb{R}$, the LASSO algorithm solves the following optimization problem:

$$\widehat{\beta} = \underset{\beta}{\operatorname{argmin}} \left\{ \frac{1}{2n} \sum_{i=1}^{n} \left(y_i - x_i^T \beta\right)^2 + \lambda \parallel \beta \parallel_1 \right\}$$

where the vector $\beta \in \mathbb{R}^p$ is a regression coefficient, the regularization of the L_i norm $\parallel \beta \parallel_1 = \sum_{j=1}^{p} |\beta_j|$ induces sparsity in the solution, and $\lambda \geq 0$ is a regularization parameter that controls the amount of shrinkage: Larger values of λ result in greater regularization and a sparser coefficient vector.

The parameter λ is crucial in the LASSO method. A λ of zero reduces the LASSO regression to a simple linear regression, whereas very large values of λ will result in all coefficient estimates approaching zero.

Random forest. The random forest (RF) algorithm is an ensemble learning method that constructs a multitude

of decision trees during training and outputs the class that is the mode (for classification) or the mean prediction (for regression) of the individual trees for a given input. Given a dataset D with n samples, the RF algorithm operates as follows:

For $b = 1$ to B (where B is the number of trees to build):

1. Draw a bootstrap sample D_b of size n from the dataset D with replacement.
2. Grow a decision tree T_b from the bootstrap sample D_b. At each node
   a. Select m variables randomly from all variables.
   b. Pick the best variable/split-point among the m.
   c. Split the node into 2 child nodes.

   Output the ensemble of trees $\{T_1, T_2, \ldots T_B\}$.

   For a new input sample x:

- Classification: The output class is the mode of the classes output by $\{T_1, T_2, \ldots T_B\}$.

$$\widehat{\mathrm{y}}(\mathrm{x}) = \mathrm{mode}(\mathrm{T}_1(\mathrm{x}), \mathrm{T}_2(\mathrm{x}), \ldots \mathrm{T}_\mathrm{B}(\mathrm{x}))$$

- Regression: The output is the mean of the outputs by $\{T_1, T_2, \ldots T_B\}$.

$$\widehat{\mathrm{y}}(\mathrm{x}) = \frac{1}{\mathrm{B}} \sum_{\mathrm{b}=1}^{\mathrm{B}} \mathrm{T}_\mathrm{b}(\mathrm{x})$$

The parameter B denotes the number of trees in the forest, and an increase in B can lead to improved accuracy, albeit possibly resulting in a more complex and slower model. On the other hand, m represents the number of variables considered for splitting at each node. For classification tasks, m is typically set to approximately $\sqrt{p}$, whereas for regression, it is around $p/3$, with p being the total number of input variables.

Extreme gradient boosting. eXtreme Gradient Boosting (XGBoost) is a gradient-boosting framework that builds an ensemble of weak prediction models, typically decision trees. Given a differentiable loss function $L(y, \hat{y})$, where y is the true output and $\hat{y}$ is the predicted output, the objective function to be optimized in the case of regression problems can be expressed as

$$\mathrm{Obj} = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{L}(\mathrm{y}_\mathrm{i}, \widehat{\mathrm{y}}_\mathrm{i}) + \sum_{\mathrm{k}=1}^{\mathrm{K}} \Omega(\mathrm{f}_\mathrm{k})$$

where $\Omega(f_k)$ is the regularization term on the kth tree and K is the total number of trees. The regularization term penalizes the complexity of the model, preventing overfitting.

For tree ensembles, the regularization term is defined as

$$\Omega(\mathrm{f}) = \gamma \mathrm{T} + \frac{1}{2} \lambda \sum_{\mathrm{j}=1}^{\mathrm{T}} \omega_\mathrm{j}^2$$

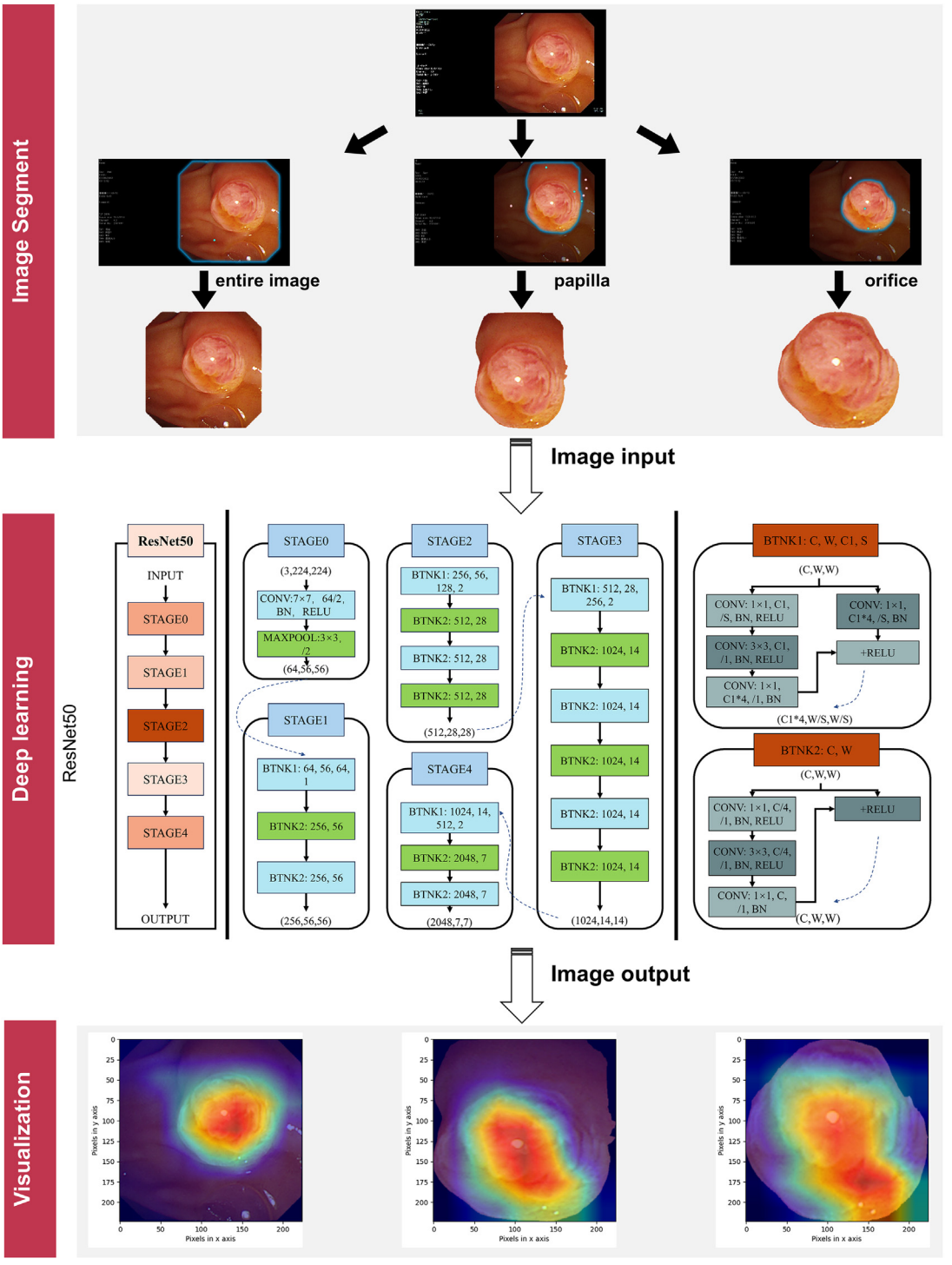
where T is the number of leaves in the tree, ω_j is the score assigned to the jth leaf, γ penalizes the number of leaves, and λ regularizes the leaf scores.

The XGBoost algorithm has a multitude of parameters. Among the most significant are the following:

- learning_rate (or η): Step size at each iteration while moving toward a minimum of the loss function. Lower values make the optimization more robust.
- max_depth: Maximum depth of the decision tree. Higher values can capture more complex patterns but risk overfitting.
- n_estimators: Number of boosting rounds or trees added to the model.
- subsample: Fraction of training data to sample in each boosting round. This can prevent overfitting.
- γ: Regularization parameter. Controls the number of leaves in the trees.
- λ: Regularization parameter of L2 regularization term on the weights. Used to avoid overfitting.

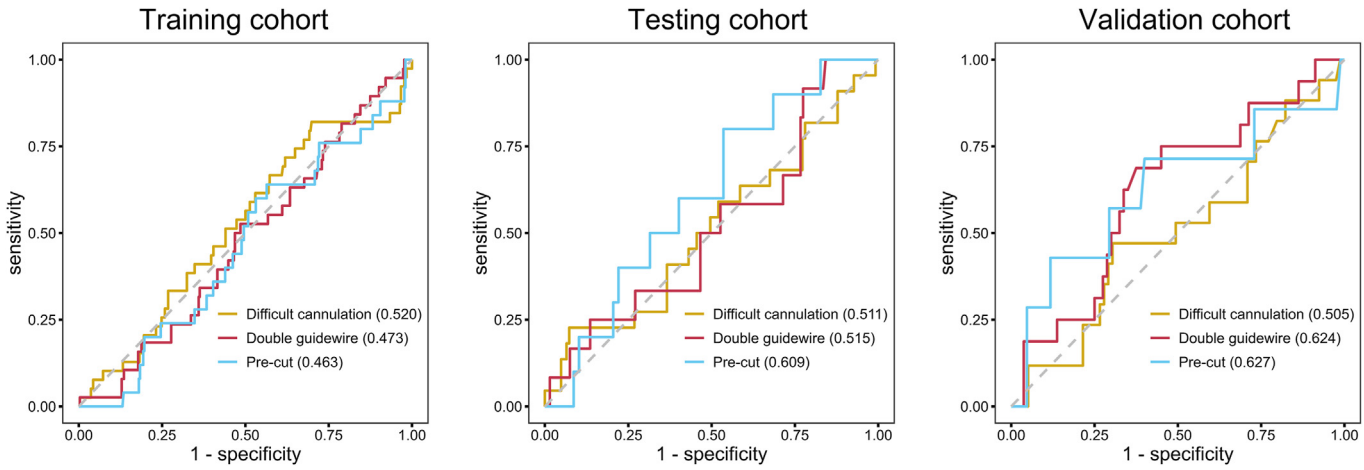
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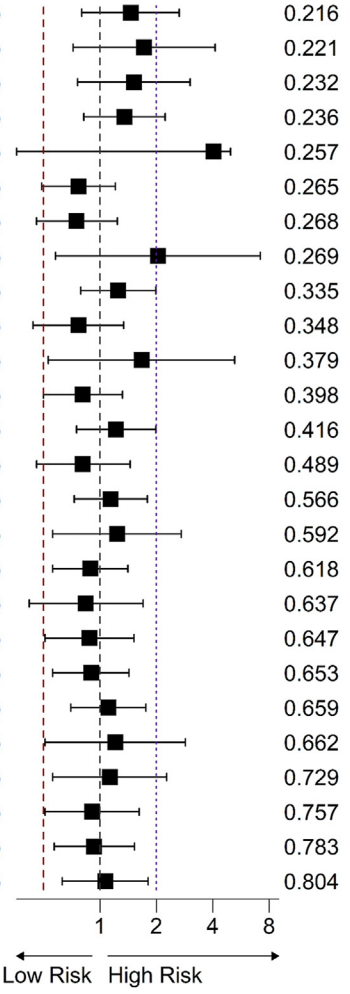
CONV: Convolution, representing a convolutional layer.
BN: Batch Normalization, used to standardize the output of each layer, helping to accelerate training and stabilize the model.
RELU: Rectified Linear Unit, an activation function defined as f(x) = max(0, x).
MAXPOOL: Max Pooling, a pooling operation used to downsample the spatial dimensions.
BTNK: Bottleneck, a bottleneck block in ResNet, designed to reduce computation and improve training efficiency.
C: Channels, indicating the number of convolutional filters (output channels).
W: Width, representing the width of the feature map.
S: Stride, the step size with which the convolution or pooling operation moves.

Supplementary Figure 1. Segmentation of papillary images and the deep learning network to predict the probability of post-ERCP pancreatitis through extracting radiomic features from corresponding endoscopic images.

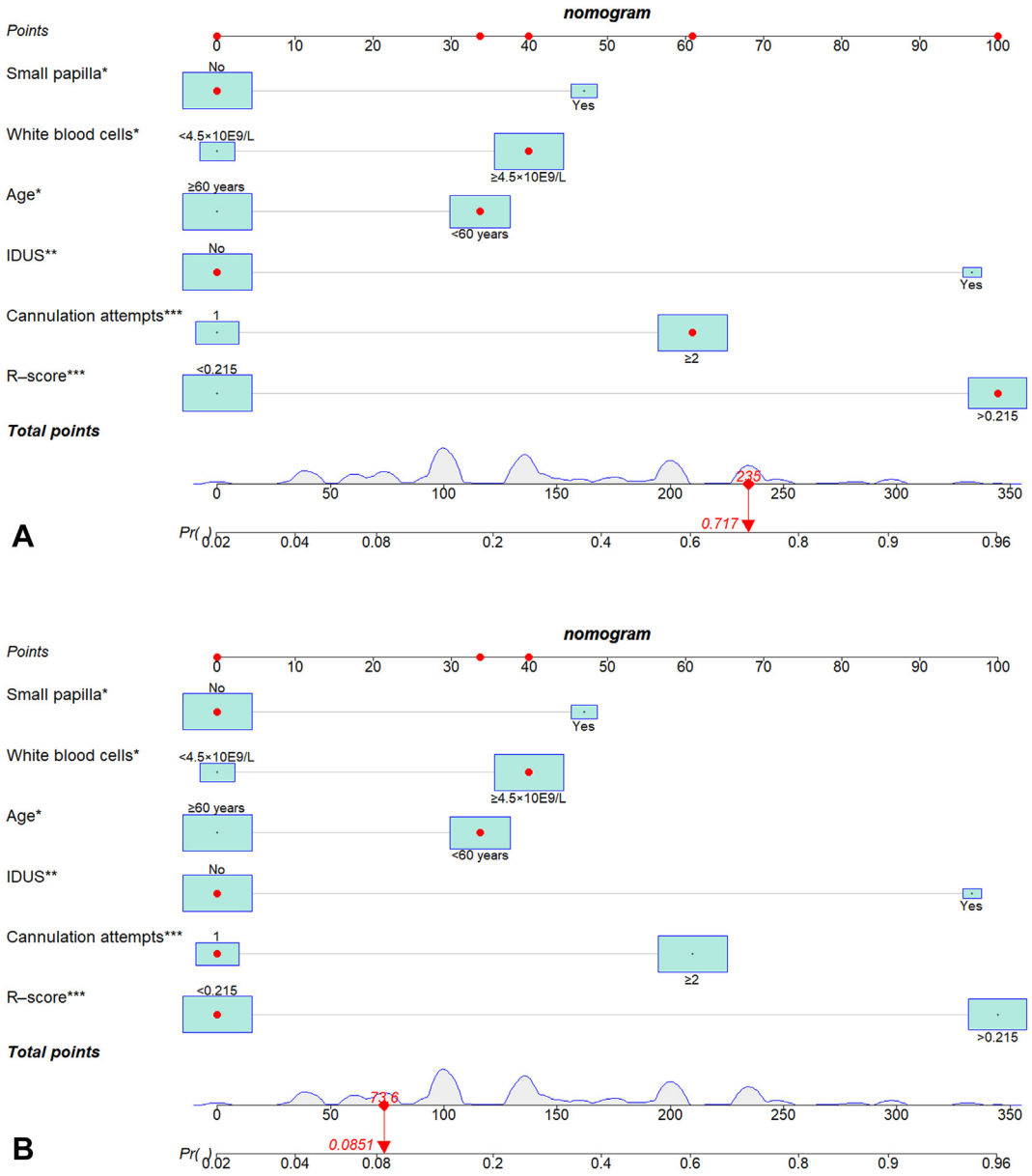


Supplementary Figure 2. Receiver-operating characteristic curves of the radiomic score for difficult cannulation, guidewire technique, and need for precut sphincterotomy in the training, testing, and validation cohorts.

Variable	non-PEP (n=227) (Events/Events)	PEP (n=114) (Events/Events)	OR (95%CI)	p value
Biliary Brushing (no vs. yes)	32/195	22/92	1.46 (0.80-2.65)	0.216
Biliary Biopsy (no vs. yes)	12/215	10/104	1.72 (0.72-4.12)	0.221
Cannulation methods (single vs. double)	205/22	98/16	1.52 (0.76-3.03)	0.232
Prothrombin time (<12.35 vs. ≥12.35 s)	171/56	79/35	1.35 (0.82-2.23)	0.236
Autoimmune diseases (no vs. yes)	226/1	112/2	4.04 (0.36-4.98)	0.257
Fibrinogen (<3.41 vs. ≥3.41 g/L)	99/128	57/57	0.77 (0.49-1.21)	0.265
AST (<123 vs. ≥123 U/L)	157/73	84/30	0.75 (0.46-1.24)	0.268
Single-operator cholangioscopy (no vs. yes)	222/5	109/5	2.04 (0.58-7.18)	0.269
Cholinesterase (<6909 vs. ≥6909 U/L)	98/129	43/71	1.25 (0.79-1.99)	0.335
Balloon extraction (no vs. yes)	173/54	92/22	0.77 (0.44-1.34)	0.348
Maximal CBD diameter (<0.55 vs. >0.55 cm)	13/214	4/110	1.67 (0.53-5.24)	0.379
Pancreatic sphincterotomy (no vs. yes)	151/76	114/33	0.81 (0.50-1.32)	0.398
INR (<1.04 vs. ≥1.04 U/L)	163/64	77/37	1.22 (0.75-1.99)	0.416
Alcohol drink (no vs. yes)	180/47	94/20	0.81 (0.46-1.45)	0.489
Platelets (<216.5 vs. >216.5 10E9/L)	117/110	55/59	1.14 (0.73-1.79)	0.566
Pancreatic duct dilation (no vs. yes)	209/18	103/11	1.24 (0.56-2.72)	0.592
Albumin (<42.1 vs. ≥42.1 g/L)	133/94	70/44	0.89 (0.56-1.41)	0.618
Trainee involvement (no vs. yes)	24/203	14/100	0.84 (0.42-1.70)	0.637
Precut sphincterotomy (no vs. yes)	214/13	102/12	0.88 (0.51-1.52)	0.647
ALT (<47 vs. ≥47 U/L)	80/147	43/71	0.90 (0.56-1.43)	0.653
Sex (female vs. male)	86/141	46/68	1.11 (0.70-1.76)	0.659
Diabetes (no vs. yes)	212/15	105/9	1.21 (0.51-2.86)	0.662
Difficult cannulation (no vs. yes)	202/25	100/14	1.13 (0.56-2.27)	0.729
Periampullary diverticulum (no vs. yes)	182/45	93/21	0.91 (0.51-1.62)	0.757
Hypertension (no vs. yes)	158/69	81/33	0.93 (0.57-1.53)	0.783
Smoke (no vs. yes)	174/53	86/28	1.07 (0.63-1.81)	0.804



Supplementary Figure 3. Forest plot presenting ORs, CIs, and *P* values for unselected variables by the univariate analysis. *OR*, Odds ratio; *CI*, confidence interval; *PEP*, post-ERCP pancreatitis; *AST*, aspartate aminotransferase; *CBD*, common bile duct; *INR*, International normal ratio; *ALT*, alanine aminotransferase.



Supplementary Figure 4. Representative nomograms of high-risk (**A**) and low-risk (**B**) patients for post-ERCP pancreatitis. *IDUS*, Intraductal US; *R-score*, radiomic score. *$P < .05$, **$P < .01$, ***$P < .001$.

SUPPLEMENTARY TABLE 1. Definitions, cutoffs, and scores of included factors

Patient-related factors	Cutoff	Score	Procedure-related factors	Cutoff	Score	Radiomics of the papilla	Cutoff	Score
Sex	Female	0	ERCP type	Diagnostic	0	Radiomic score	<.215	0
	Male	1		Therapeutic	1		≥.215	1
Age	<60 y	1	Cannulation time,* min	<1	0			
	≥60 y	0		≥1	1			
Body mass index	<28.5 kg/m^2	0	Cannulation attempts†	1	0			
	≥28.5 kg/m^2	1		≥2	1			
Pancreatic duct dilation‡	No	0	Cannulation methods	Single wire	0			
	Yes	1		Double wire	1			
Alanine aminotransferase	<47 U/L	0	Difficult cannulation§	No	0			
	≥47 U/L	1		Yes	1			
Aspartate aminotransferase	<123 U/L	0	Inadvertent pancreatic duct cannulation	No	0			
	≥123 U/L	1		Yes	1			
γ-Glutamyl transferase	<29.5 g/L	0	Pancreatic deep-wire implantation	No	0			
	≥29.5 g/L	1		Yes	1			
Albumin	<42.1 g/L	0	Major papilla sphincterotomy	No	0			
	≥42.1 g/L	1		Yes	1			
Total bilirubin	<156 μmol/L	0	Endoscopic papillary balloon dilatation	No	1			
	≥156 μmol/L	1		Yes	0			
Direct bilirubin	<7.5 μmol/L	0	Pancreatic opacification	No	0			
	≥7.5 μmol/L	1		Yes	1			
Serum calcium	<2.05 mmol/L	0	Balloon extraction	No	0			
	≥2.05 mmol/L	1		Yes	1			
Cholinesterase	<6909 U/L	0	Basket catheter	No	1			
	≥6909 U/L	1		Yes	0			
White blood cell count	<4.5 10^9/L	0	Biliary brushing	No	1			
	≥4.5 10^9/L	1		Yes	0			
Neutrophil ratio	<65.5	0	Biliary biopsy sampling	No	0			
	≥65.5	1		Yes	1			
Platelets	<216.5 10^9/L	0	Intraductal US	No	0			
	≥216.5 10^9/L	1		Yes	1			
Prothrombin time	<12.35 s	0	Single-operator cholangioscopy	No	0			
	≥12.35 s	1		Yes	1			
Fibrinogen	<3.41 g/L	0	Endoscopic nasobiliary drainage	No	0			
	≥3.41 g/L	1		Yes	1			
International normal ratio	<1.04	0	Trainee involvement	No	0			
	≥1.04	1		Yes	1			
Hypertension	No	1	Stent types	No + plastic stent	0			
	Yes	0		Multiple plastic stents	1			

(*continued on the next page*)

SUPPLEMENTARY TABLE 1. Continued

Patient-related factors	Cutoff	Score	Procedure-related factors	Cutoff	Score	Radiomics of the papilla	Cutoff	Score
Diabetes	No	0	Precut methods	No	0			
	Yes	1		Yes	1			
Liver cirrhosis	No	0						
	Yes	1						
Autoimmune diseases	No	0						
	Yes	1						
Alcohol use	No	0						
	Yes	1						
Smoker	No	0						
	Yes	1						
Papilla types‖	Types 1+3+4	0						
	Type 2	1						
Periampullary diverticulum	No	0						
	Yes	1						
Maximal common bile duct diameter	<.55 cm	0						
	≥.55 cm	1						
Distal common bile duct diameter‖¶	<.75 cm	0						
	>.75 cm	1						

*The time between the first attempt of cannulation and secure placement of the guidewire in the common bile duct.
†An attempt was defined as sustained contact between the cannulation device and the papilla for ≥5 seconds (Bailey AA, Bourke MJ, Kaffes AJ, et al. Needle-knife sphincterotomy: factors predicting its use and the relationship with post-ERCP pancreatitis (with video). Gastrointest Endosc 2010;71:266-71).
‡This was defined as ≥5 mm in diameter (Umemura M, Shimura E, Asai Y, et al. Dilated main pancreatic duct can be a negative predictor of pancreatitis related to biliary SEMS insertion across the papilla. Scand J Gastroenterol 2021;56:865-69).
§Cannulation attempts ≥10.
‖Types were classified according to Haraldsson's classification: type 1, regular; type 2, small; type 3, protruding or pendulous; type 4, creased or ridged (Haraldsson E, Kylänpää L, Grönroos J, et al. Macroscopic appearance of the major duodenal papilla influences bile duct cannulation: a prospective multicenter study by the Scandinavian Association for Digestive Endoscopy Study Group for ERCP. Gastrointest Endosc 2019;90:957-63).
¶The 1-cm proximal portion from the opening of duodenal major papilla (Huang L, Xu Y, Chen J, et al. An artificial intelligence difficulty scoring system for stone removal during ERCP: a prospective validation. Endoscopy 2023;55:4-11.)

SUPPLEMENTARY TABLE 2. Deep learning of different regions of the papilla with different approaches in the derivation cohort

Model	Region	Cutoff	Area under the curve	Accuracy	Sensitivity	Specificity	*P* value*
ResNet18 (Microsoft Research)	Orifice	.928	.641 (.520-.762)	63.2 (62.7-63.7)	67.6 (52.5-82.7)	60.0 (46.4-73.6)	.109
	Papilla	.984	.630 (.506-.754)	69.0 (68.5-69.4)	37.8 (22.2-53.5)	92.0 (84.5-99.5)	.062
	Entire image	.041	.596 (.474-.718)	60.9 (60.4-61.5)	62.2 (46.5-77.8)	60.0 (46.4-73.6)	.034
ResNet34 (Microsoft Research)	Orifice	.082	.582 (.456-.707)	64.4 (63.9-64.9)	37.8 (22.2-53.5)	84.0 (73.8-94.2)	.006
	Papilla	.010	.629 (.510-.749)	62.1 (61.5-62.6)	73.0 (58.7-87.3)	54.0 (40.2-67.8)	.072
	Entire image	.034	.465 (.340-.591)	58.6 (58.1-59.2)	21.6 (8.4-34.9)	86.0 (76.4-95.6)	.007
ResNet50 (Microsoft Research)	Orifice	.315	.524 (.398-.650)	60.9 (60.4-61.5)	40.5 (24.7-56.4)	76.0 (64.2-87.8)	.005
	Papilla	.215	.759 (.658-.861)	73.6 (73.1-74.0)	81.1 (68.5-93.7)	68.0 (55.1-80.9)	NA
	Entire image	.250	.539 (.414-.664)	59.8 (59.2-60.3)	43.2 (27.3-59.2)	72.0 (59.6-84.4)	<.001
ResNet101 (Microsoft Research)	Orifice	.339	.496 (.373-.619)	52.9 (52.3-53.4)	75.7 (61.9-89.5)	36.0 (22.7-49.3)	<.001
	Papilla	.243	.517 (.394-.640)	51.7 (51.2-52.3)	89.2 (79.2-99.2)	24.0 (12.2-35.8)	<.001
	Entire image	.017	.497 (.373-.621)	47.1 (46.6-47.7)	99.8 (99.5-100.0)	8.0 (.5-15.5)	.002

NA, not applicable.
Values in parentheses are 95% confidence intervals.
*Compared with the area under the curve of the radiomic score of the papilla by ResNet50.

SUPPLEMENTARY TABLE 3. Overview of patients with moderate to severe pancreatitis

	Derivation cohort			Validation cohort		
	Mild (n = 140)	Moderate to severe (n = 22)	*P* value	Mild (n = 26)	Moderate to severe (n = 6)	*P* value
Pancreatitis, %	86.42	13.58		81.25	18.75	
Average hospital duration, days	2.79 ± .87	8.05 ± 5.43	<.001	2.88 ± .95	34.50 ± 39.26	<.001
Postoperative amylase level, U/L	1334.9 ± 525.4	1693.28 ± 556.66	.009	1367.32 ± 559.66	2,000.83 ± 303.07	.012
Mortality, n (%)	0 (0)	5 (22.7)	<.001	0 (0)	1 (16.7)	<.001

SUPPLEMENTARY TABLE 4. Performance of R-score and the predictive model for moderate to severe post-ERCP pancreatitis

	Area under the curve (95% confidence interval)	Accuracy	Sensitivity	Specificity
Derivation cohort				
R-score	.697 (.604-.789)	.628	.773	.621
Predictive model	.775 (.603-.799)	.730	.727	.731
Validation cohort				
R-score	.594 (.381-.808)	.531	.667	.522
Predictive model	.780 (.559-.960)	.708	.833	.700

R-score, Radiomic score.